The Shifting Lexicon of American Power: Computational Analysis of Political Language in State of the Union Addresses, 1950–2026

Calder Katyal
Yale University
HUMS 3210 (Culture as Data: Thinking Computationally with Text)
calder.katyal@yale.edu

Spring 2026

Abstract

This paper applies computational text analysis to a corpus of 79 State of the Union addresses delivered between 1950 and 2026, spanning fifteen presidencies from Truman through Trump's second term. Using TF-IDF differential analysis, Latent Dirichlet Allocation topic modeling, supervised classification, and t-SNE document embeddings, we find that (1) SOTU vocabulary has transformed substantially across eras, tracking geopolitical and domestic pressures with remarkable fidelity; (2) Democratic and Republican presidents employ measurably different rhetorical vocabularies, with a Linear SVM classifier achieving 92.4% cross-validated accuracy in distinguishing party from text alone; and (3) temporal proximity is at least as powerful a structuring force as party affiliation in the embedding space. The central argument is that the language of the State of the Union is shaped more by the historical moment a president inhabits than by the party to which the president belongs—a finding with implications for how we understand partisan polarization through text.

1 Introduction

The State of the Union address occupies a peculiar position in the archive of American political language. Mandated by Article II, Section 3 of the Constitution, it is one of the few textual artifacts produced under nearly identical institutional constraints across decades: a single speaker (the president), a fixed audience (Congress and, since the television era, the public), a recurring occasion, a shared rhetorical purpose. These constraints make the SOTU corpus an unusually controlled object for studying how political language changes—and what those changes reveal about the relationship between rhetoric and historical context.

This paper asks three questions. First, how does the language of SOTU addresses change across the period 1950–2026? Second, do Democratic and Republican presidents use detectably different vocabularies? Third, how well can the methods of computational text analysis—what Moretti (Moretti, 2013) calls "distant reading"—capture these patterns, and what do those methods fail to see?

The approach draws on a range of methods: vector semantics (TF-IDF), generative topic modeling (LDA), supervised machine learning, and dimensionality reduction for visualization. Following Underwood's (2019) argument that computational methods can surface patterns "invisible to human readers operating at the scale of individual texts," we treat each address as a document in a corpus and apply distributional methods to detect temporal drift and partisan divergence simultaneously. But we remain attentive to the cautions of Drucker (2017) and D'Ignazio and Klein (2020), who remind us that data visualization and quantitative modeling encode assumptions—about what is counted, what is comparable, and what is rendered invisible in the act of abstraction.

2 Data Curation and Preprocessing

2.1 Assembling the Corpus

The construction of any corpus is an interpretive act, and it is worth describing our choices in some detail. The corpus consists of 79 State of the Union addresses delivered by 15 presidents between 1950 and 2026. Speeches from 1950 through 2020 (73 texts) were sourced from the `stdlib-js/datasets-sotu` public repository on GitHub, which itself draws on the authoritative collection maintained by the American Presidency Project at UC Santa Barbara. Speeches from 2021 through 2026—Biden's four addresses and Trump's two second-term addresses—were collected directly from the American Presidency Project (`presidency.ucsb.edu`), where each address is published as a full transcript with speaker annotations and audience reactions noted inline.

This two-source approach introduces a minor inconsistency: the stdlib-js texts are lightly cleaned, while the Presidency Project transcripts retain stage directions (e.g., "[Applause]"). We retained these artifacts rather than impose retroactive normalization, on the principle that cleaning decisions risk erasing information about the performative context of speech. The practical effect is negligible—stage directions constitute fewer than 2% of tokens, and our document-frequency thresholds filter most of them.

The decision to begin in 1950 reflects both pragmatic and analytical considerations. Before the mid-twentieth century, SOTU addresses were almost exclusively written messages sent to Congress—a fundamentally different genre from the televised oral address that became standard after Truman. Our corpus captures a relatively coherent rhetorical form, even as that form evolves from radio-era formality to the applause-punctuated cadences of contemporary addresses.

Table 1 summarizes the corpus by president.

The corpus contains 37 Democratic speeches and 42 Republican speeches—a modest imbalance reflecting the distribution of presidential terms across the period. The average speech length is approximately 6,014 words, though with substantial variation: Carter's 1981 written message exceeds 33,000 words, while Nixon's 1973 message—a brief, formally written report submitted at the start of his second term—falls below 1,700. This variance is itself analytically significant, as it indexes the persistence of written policy documents alongside performed oratory across the postwar period.

Table 1: SOTU Corpus Summary by President

President	**Party**	**Years**	N
Truman	D	1950–53	4
Eisenhower	R	1953–61	9
Kennedy	D	1961–63	3
Johnson	D	1964–69	6
Nixon	R	1970–74	5
Ford	R	1975–77	3
Carter	D	1978–81	4
Reagan	R	1982–88	7
G.H.W. Bush	R	1989–92	4
Clinton	D	1993–2000	8
G.W. Bush	R	2001–08	8
Obama	D	2009–16	8
Trump (1st)	R	2017–20	4
Biden	D	2021–24	4
Trump (2nd)	R	2025–26	2
Total			**79**

2.2 Preprocessing Pipeline

All texts were tokenized using NLTK's `word_tokenize` function and lowercased. For content-word analyses, stopwords were removed using the NLTK English stopword list. For TF-IDF and topic modeling, we retained both unigrams and bigrams, applied a minimum document frequency of 3 and a maximum document frequency threshold of 90%, and limited the vocabulary to 5,000 features. These parameters follow standard practice in computational text analysis (Jockers, 2013), balancing vocabulary coverage against noise. Sentence-level statistics (average sentence length, type-token ratio) were computed using NLTK's `sent_tokenize` prior to stopword removal, preserving the syntactic structure of the original text.

3 Methods and Results

3.1 Lexical Trends Over Time

We begin with what might be called the material history of the SOTU address: how its length and lexical texture have changed over seven decades.

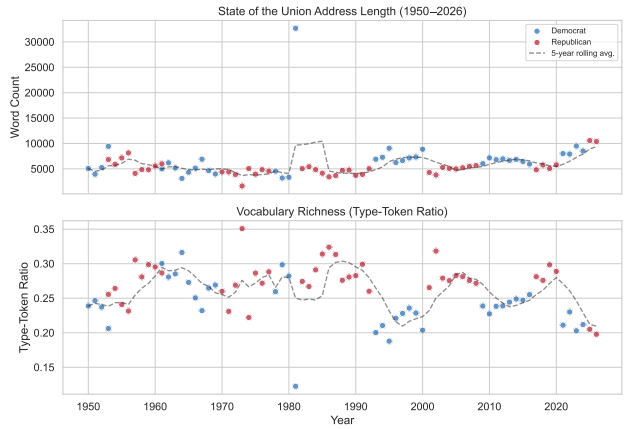

Figure 1: Speech length (word count) and type-token ratio, 1950–2026. Points colored by party; dashed line shows a 5-year rolling average.

Figure 1 reveals two patterns. Speech length oscillates rather than declines: most decades cluster between 4,000 and 7,000 words, the 1970s mark a trough ($\approx$ 4,100), and the 1980s mean is inflated by Carter's 33,558-word written message of 1981, delivered after he had lost re-election. The striking departure comes at the end of the period: Biden's later addresses (2023–2024) and Trump's two second-term addresses (2025–2026) are among the longest in the corpus—ranging from roughly 8,500 to 10,700 words—and the 2020s mean of 8,760 words is the highest of any decade. (This expansion is too large to be attributed to the inclusion of stage directions in post-2020 transcripts, which constitute well under 1% of tokens.) Second, contrary to a familiar narrative about "richer" modern political speech, the type-token ratio does *not* rise across the period. After peaking in the 1960s–80s (mean TTR $\approx$ 0.275), vocabulary diversity drops in the 1990s, recovers somewhat in the 2000s, and reaches its lowest values in the 2020s ($\approx$ 0.221). Aggregated across the full corpus, Republican speeches exhibit higher type-token ratios than Democratic ones (0.277 vs. 0.240), though this pattern is not uniform across decades—in the 1970s and the 2020s, Democratic and Republican TTRs are roughly comparable.

3.2 Partisan Language: Differential TF-IDF

To identify the linguistic signatures of each party, we computed TF-IDF vectors for each speech individually and then averaged the resulting feature vectors by party—a per-speech averaging strategy that prevents the longer Carter and Truman messages from dominating the aggregate signal. Figure 2 shows the top 20 differentially weighted terms for each party.

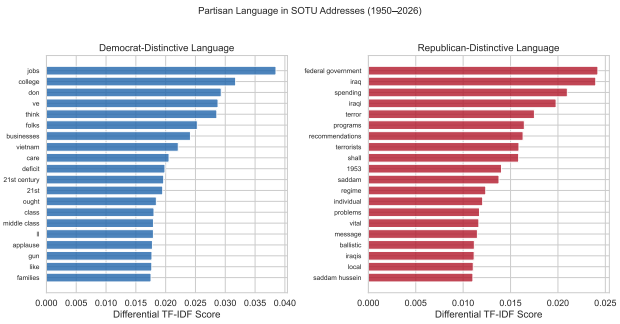

Figure 2: Top 20 terms by differential TF-IDF score for each party. Higher scores indicate stronger association with that party's speeches.

The results are immediately legible as partisan vocabularies. Republican-distinctive terms include *iraq*, *federal government*, *spending*, *terror*, and *terrorists*—a lexicon shaped heavily by the George W. Bush era and the party's rhetorical emphasis on limited government and national security. Democratic-distinctive terms include *jobs*, *college*, *middle class*, *businesses*, *folks*, and *21st century*—reflecting a focus on economic mobility and an informal, direct-address rhetorical mode prominent in the Clinton, Obama, and Biden eras.

But these signatures are not purely ideological. They are also, crucially, *temporal*. The strong Republican association with "Iraq" and "terror" reflects the historical fact that the longest continuous stretch of Republican SOTU addresses (2001–2008) coincided with the post-9/11 period. The Democratic association with colloquialisms like "folks" may say less about ideology than about the evolution of presidential speech toward conversational registers since the 1990s. This entanglement of

party and period is a recurring methodological challenge—one that structures the remainder of our analysis.

3.3 Topic Modeling (LDA)

We applied Latent Dirichlet Allocation with $k = 8$ topics to the full corpus, following Chang et al.'s (2009) recommendation to label topics by their top words rather than imposing interpretive labels post hoc.

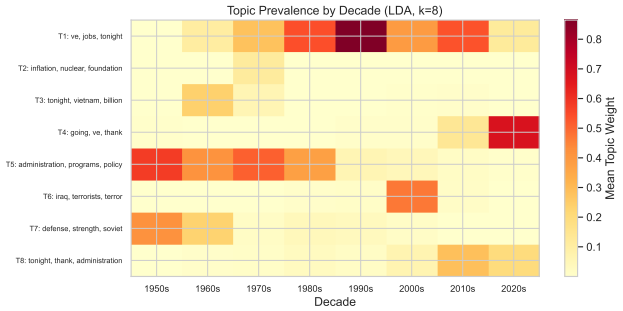

Figure 3: Mean topic prevalence by decade (LDA, $k = 8$). Rows are topics labeled by top-3 words; columns are decades.

Figure 3 displays topic prevalence by decade as a heatmap. The temporal structure is striking. A topic dominated by *defense, strength, soviet* concentrates in the 1950s and 1960s and fades sharply thereafter, once the early Cold War lexicon loses much of its referential grip. A topic centered on *tonight, vietnam, billion* peaks in the 1960s and tapers off in the 1970s. The *iraq, terrorists, terror* topic is sharply localized to the 2000s. A bureaucratic topic of *administration, programs, policy* dominates the mid-twentieth-century addresses (with weights between roughly 0.4 and 0.6 from the 1950s through the 1980s) and collapses in the 1990s. Meanwhile, a topic featuring *going, ve, thank* is essentially absent before the 2010s and rises sharply afterward, dominating the 2020s—capturing the shift toward conversational, audience-interactive rhetoric in the most recent SOTUs.

The overall trajectory amounts to a stratigraphy of American political language: Cold War defense rhetoric gives way to Great Society programs, then post-Vietnam economic anxiety, the War on Terror, and finally the colloquial populism of the 2010s–2020s. As Nelson (2010) argues, topic models work best as "starting points for interpretation"—guides to where close reading should begin.

3.4 Supervised Classification

Can a machine learning classifier predict the president's party from the text of a SOTU address? We tested four classifiers using 5-fold stratified cross-validation on TF-IDF features (3,000 features, unigrams and bigrams). Table 2 reports the results.

Table 2: Party Classification Accuracy (5-Fold CV)

Model	Mean Acc.	Std.
Logistic Regression	0.798	0.091
Linear SVM	**0.924**	0.061
Random Forest	0.823	0.060
Gradient Boosting	0.708	0.081

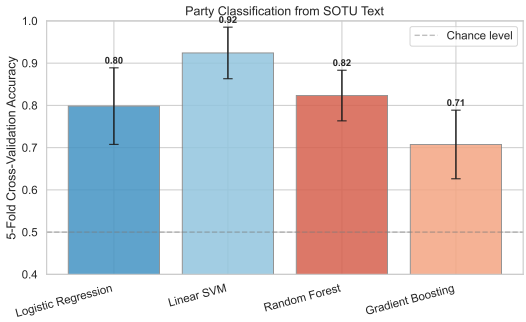

Figure 4: Classification accuracy across four models. Error bars show standard deviation across folds. Dashed line indicates the majority-class baseline (53%, given 42/79 Republican speeches).

The Linear SVM achieves 92.4% accuracy, substantially above the 53% majority-class baseline. That a linear classifier operating on bag-of-words features can recover party affiliation at this rate is itself a non-trivial finding: it suggests that partisan vocabulary differences are not only real but largely separable in a high-dimensional sparse representation. This is consistent with So's (2017) observation that even imperfect models can be informative when their behavior is interpretable.

3.5 Document Embedding Visualization

To examine the global structure of the corpus without the mediation of a classifier, we projected all 79 speeches into two dimensions using TF-IDF features, dimensionality reduction via Truncated SVD (30 components), and t-SNE (perplexity=15).

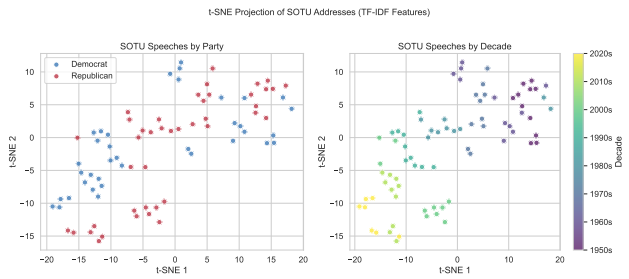

Figure 5: t-SNE projection of SOTU addresses. Left: colored by party. Right: colored by decade.

Figure 5 is perhaps the most revealing visualization in this paper. The left panel, colored by party, shows meaningful but imperfect clustering: Democratic and Republican speeches tend to occupy different regions of the space, but with substantial overlap. The right panel, colored by decade, tells a different story. Here, the chronological gradient is unmistakable—1950s speeches in one region, 2020s speeches in another, with a smooth temporal progression between. A Clinton speech sits closer to a George W. Bush speech than to a Truman speech, because the shared linguistic context of the 1990s–2000s exerts a stronger gravitational pull than party affiliation.

This aligns with Underwood's (2019) finding that historical period routinely outweighs other categorical distinctions in large-scale text analysis—and implies that models trained to detect partisanship may inadvertently be detecting periodization.

3.6 Keyword Trends and Sentence Complexity

We tracked the per-speech frequency of terms related to four thematic categories—security and defense, economy, social policy, and foreign policy—across the full corpus (Figure 6).

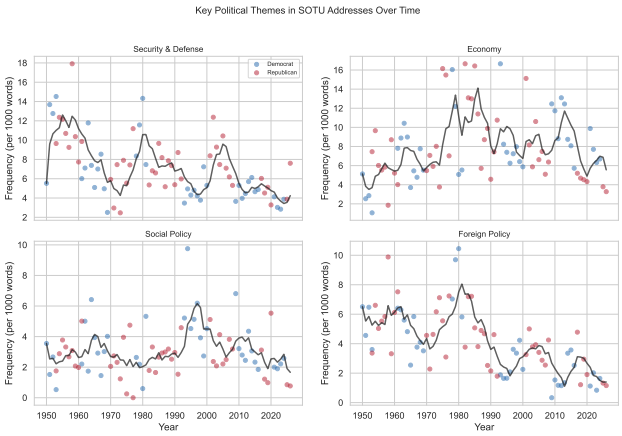

Figure 6: Frequency of key political terms (per 1,000 words) across four thematic domains, 1950–2026.

The trends are historically interpretable. Security and defense language reaches its highest values in the 1950s (≈ 11.6 per 1,000 words)—driven by the Korean War and early Cold War—drops sharply in the 1960s, registers secondary peaks during the Reagan buildup of the 1980s and the post-September-11 years, and falls to its lowest corpus-wide values in the 2010s–2020s. Economic language climbs through the 1960s and 1970s, peaks across the late-Carter and Reagan years (the 1980s mean is the highest of any decade), and remains elevated through the Clinton, Obama, and post-2008 recovery years before declining in the 2020s. Social policy terms peak in the Clinton 1990s, with secondary clusters in the George W. Bush years; the Great Society addresses, perhaps surprisingly, register only moderate values on these keyword counts. Foreign policy language is broadly stable from the 1950s through the 1980s—hovering around five mentions per 1,000 words—then drops sharply in the 1990s and continues to decline thereafter, reaching its lowest sustained values in the 2010s–2020s. This is a lexical trace of the turn toward domestic populism in both parties.

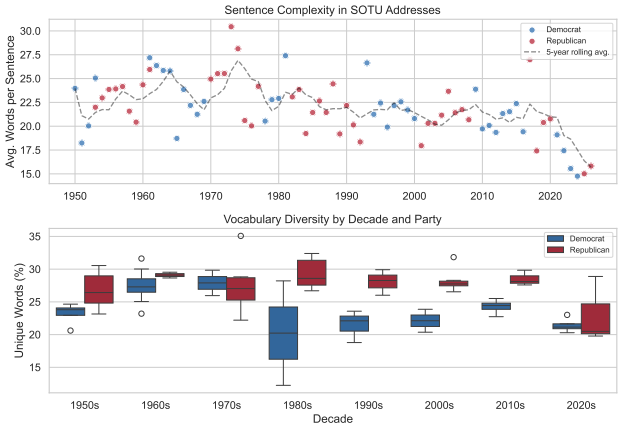

Figure 7: Sentence complexity (average words per sentence) and vocabulary diversity by decade and party.

Figure 7 examines sentence-level complexity. Average sentence length is essentially flat from the 1950s through the 1980s—hovering around 22–25 words per sentence—and then declines from the 1990s onward, falling to roughly 17 words per sentence in the 2020s. The inflection comes later than a conventional "television revolution" account would predict. The vocabulary diversity boxplots reveal a partisan pattern: Republican speeches sustain higher type-token ratios than Democratic ones in nearly every decade, with Democratic vocabulary diversity dropping noticeably from the 1980s onward.

4 Discussion

Three findings emerge from this analysis, each with implications for how we read political text computationally.

The dominance of the temporal signal. Across every method—topic modeling, keyword tracking, t-SNE, TF-IDF—historical period proved to be the primary structuring force in SOTU rhetoric. The Cold War speeches of the 1950s are linguistically distant from the post-9/11 addresses of the 2000s, which are in turn distant from the populist rhetoric of the 2020s. As Schöch (2013) argues, "big" data in the humanities is also "messy" data: the temporal signal is entangled with every other variable we might wish to isolate.

Partisan signatures are real, but historically contingent. The 92.4% classification accuracy demonstrates that Democratic and Republican presidents use detectably different language. But the nature of that difference shifts across periods. The t-SNE projection (Figure 5) suggests that 1950s–60s addresses cluster more by chronology than by party, while later speeches show somewhat greater partisan separation. Whether this represents a genuine widening of polarization or simply the entanglement of party with the distinctive lexicons of the post-9/11 and post-2020 periods is difficult to disentangle without an ablation study comparing classification on early-only versus full-corpus subsets. We flag this as a question for future work.

The limits of the bag of words. Every method in this paper reduces speeches to distributions over terms, discarding syntax, rhetorical structure, and argumentative logic—the foundational trade-off of distant reading: scale for depth. As Drucker (2017) argues, the "distant" in distant reading is not merely spatial but epistemological. The most productive use of these computational results may be to identify *where* close reading should focus: the moments where the model is most confident, or most confused.

Limitations. Our corpus, at 79 documents, is small by machine learning standards. SOTU addresses are also collaborative texts, so attributing patterns to individual presidents or parties requires caution. Future work might apply contextual embeddings such as BERT or adopt LLM-based topic evaluation methods (Yang et al., 2024).

5 Conclusion

The central finding—that historical period structures presidential language at least as powerfully as party affiliation—has implications for how we interpret claims about partisan polarization: the language of the State of the Union tracks the world the president inhabits, and party affiliation modulates this language without determining it. The most interesting result is not a classification accuracy or a topic distribution but the recognition that political language, like literary language, is shaped by forces larger than any individual speaker's intentions (Moretti, 2013; Witmore, 2012).

References

Chang, J., Boyd-Graber, J., Gerrish, S., Wang, C., and Blei, D. M. (2009). Reading tea leaves: How humans interpret topic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 288–296.

D'Ignazio, C. and Klein, L. F. (2020). Feminist data visualization. In *Data Feminism*. MIT Press, Cambridge, MA.

Drucker, J. (2017). Why distant reading isn't. In *PMLA*, volume 132, pages 628–635.

Jockers, M. L. (2013). *Macroanalysis: Digital Methods and Literary History*. University of Illinois Press, Urbana.

Moretti, F. (2013). *Distant Reading*. Verso, London.

Nelson, R. K. (2010). Mining the dispatch. *Digital Scholarship Lab, University of Richmond*.

Schöch, C. (2013). Big? smart? clean? messy? data in the humanities. In *Journal of Digital Humanities*, volume 2.

So, R. J. (2017). All models are wrong. *PMLA*, 132(3):668–673.

Underwood, T. (2019). *Distant Horizons: Digital Evidence and Literary Change*. University of Chicago Press, Chicago.

Witmore, M. (2012). Text: A massively addressable object. *Debates in the Digital Humanities*.

Yang, X., Zhao, H., Phung, D., Buntine, W., and Du, L. (2024). LLM reading tea leaves: Automatically evaluating topic models with large language models. *arXiv preprint arXiv:2406.09008*.